

Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Kanghuli, Joel
Date of birth:	9/9/1990
Subject:	030-Flight Performance&Planing
Result:	42,89 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	378	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		1,00	1,00
2	381	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		1,00	1,00
3	361	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		1,00	1,00
4	420	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		0,00	1,00
5	384	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		1,00	1,00
6	373	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		0,00	1,00
7	380	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		0,00	1,00
8	379	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		0,00	1,00
9	390	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		1,00	1,00
10	387	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		0,00	1,00
11	391	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		1,00	1,00
12	376	31. MASS AND BALANCE - AEROPLANES/HELICOPTERS		0,00	1,00
13	183	31.2 LOADING		1,00	1,00
14	257	31.3.1 Definition of centre of gravity		1,00	1,00
15	278	31.3.3 Basic calculations of CG		0,00	2,00
16	404	32. PERFORMANCE - AEROPLANES		0,00	1,00
17	425	32. PERFORMANCE - AEROPLANES		1,00	1,00
18	118	32.2.2 Effect of Variables on Single-Engine Aeroplane Performance		1,00	1,00
19	87	32.2.5.1 Take-off		0,00	1,00
20	61	32.2.5.1 Take-off		0,00	1,00
21	36	32.2.5.4 Landing		1,00	1,00
22	75	32.2.5.4 Landing		1,00	1,00
23	31	32.4.1 Take - off		1,00	1,00
24	197	32.4.1.7 Take-off climb		0,00	1,00
25	82	32.4.3 Cruise		1,00	1,00
26	202	32.4.3 Cruise		0,00	1,00
27	129	32.4.3 Cruise		0,00	1,00
28	117	32.4.3.4 Long Range Cruise		0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	24	32.4.3.5	Influence of variables on cruise performance	1,00	1,00
30	21	32.4.5.1	Descent techniques	1,00	1,00
31	418	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
32	413	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
33	414	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
34	421	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
35	417	33.1	FLIGHT PLANNING FOR VFR FLIGHTS	0,00	1,00
36	419	33.1	FLIGHT PLANNING FOR VFR FLIGHTS	0,00	1,00
37	148	33.1.1.1	Routes, airfields, heights and altitudes from VFR charts	0,00	1,00
38	259	33.1.1.5	Completion of navigation plan	0,00	2,00
39	2	33.2.1	IFR Navigation plan	0,00	1,00
40	88	33.5.1.1	Format of Flight Plan	0,00	1,00
Total:43%				18,00	42,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: Kanghuli, Joel

Location: Uganda Civil Aviation Authority

Your Result: 18,00 from 42,00 Points (42,86 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	15	8,00	16,00	50,00
31.2. LOADING	1	1,00	1,00	100,00
31.3. FUNDAMENTALS OF CG CALCULATIONS	2	1,00	3,00	33,33
32. PERFORMANCE - AEROPLANES	15	8,00	15,00	53,33
32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES	5	3,00	5,00	60,00
32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY	8	4,00	8,00	50,00
33. FLIGHT PLANNING AND FLIGHT MONITORING	10	2,00	11,00	18,18
33.1. FLIGHT PLANNING FOR VFR FLIGHTS	4	0,00	5,00	0,00
33.2. FLIGHT PLANNING FOR IFR FLIGHTS	1	0,00	1,00	0,00
33.5. ICAO FLIGHT PLAN (ATS Flight Plan)	1	0,00	1,00	0,00
Total	40	18,00	42,00	42,86

Candidate: Joel Kanghuli
Date of birth: 1990-09-09

Subject: 030-Flight Performance&Planing
Date: 2014-07-18



1

What is the location of the CG if 60 pounds are removed from Station 70?

Aircraft weight 8,420 lb

CG location Station 85

☐ 84.9.

☐ 84.1.

☒ 85.1.

2

Could 100 pounds of weight be shifted from Station 30.0 to Station 120.0 without exceeding the aft CG limit?

Total weight 4,750 lb

CG location Station 115.8

☐ Yes; the CG would remain at Station 115.8.

☒ Yes; the new CG would be located at Station 117.69.

☐ No; the new CG would be located at Station 118.15.

3

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?
Floor load limit - 85 lb/sq ft
Pallet weight - 44 lb
Tiedown devices - 29 lb

☒ 924.5 pounds.

☐ 953.6 pounds.

☐ 968.6 pounds.

4

The center of gravity tube is

- ☐ lengthened for lighter pilots.
- ☐ shortened for lighter pilots.
- ☒ lengthened for heavier pilots.

5

An aircraft is loaded with the CG at the aft limit. What are the performance characteristics compared with the CG at the forward limit?

- ☐ The aft CG provides the highest stall speed and cruising speed.
- ☐ Cruising speed is lower because of more induced drag created by the elevator or stabilizer being required to provide more lift with an aft CG.
- ☒ The aft CG provides the lowest stalling speed, the highest cruising speed, and least stability.

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6

How much weight could be added at Station 160 without exceeding the aft CG limit?

Aircraft weight 8,300 lb

CG location Station 90.0

Aft CG limit Station 90.5

☒ 16.5 pounds.

☐ 13.9 pounds.

☐ 59.7 pounds.

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7

Could 100 pounds of weight be shifted from Station 130.0 to Station 30.0 without exceeding the forward CG limit?

Total weight 2,800 lb
CG location Station 120.0

☒ Yes; the new CG would be located at Station 117.89.

☐ No; the new CG would be located at Station 116.42.

☐ No; the new CG would be located at Station 116.89.

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8

How much weight must be shifted from Station 150.0 to Station 30.0 to move the CG to exactly the aft CG limit?

Total weight 7,500 lb

CG location Station 80.5

☒ 68.9 pounds.

☐ 62.5 pounds.

☐ 65.8 pounds.

9

Under which condition is a forward CG most critical?

☐ When in an unusual attitude.

☒ On landing.

☐ On takeoff.

10

As the CG moves aft, an aircraft becomes

- ☐ less stable, yet easier to control.
- ☐ less stable and less controllable.
- ☒ more stable and controllable as long as the aft CG is not exceeded.

11

As the CG location is changed, recovery from a stall becomes progressively

- ☐ less difficult as the CG moves rearward.
- ☐ more difficult as the CG moves either forward or rearward.
- ☒ more difficult as the CG moves rearward.

12

What is the location of the CG if 90 pounds are removed from Station 140?

Aircraft weight 6,230 lb

CG location Station 79

☒ **79.9.**

☐ **77.9.**

☐ **78.1.**

13

In flight the wing of an aircraft containing fuel is subjected to vertical loads that produce a bending moment which is:

- ☒ highest at the wing root
- ☐ equal to half the weight of the aircraft multiplied by the semi span
- ☐ equal to the zero -fuel weight multiplied by the span
- ☐ lowest at the wing root

14

When an aircraft is stationary on the ground, its total weight will act vertically

- ☐ *through the main wheels of its undercarriage assembly.*
- ☐ *through its centre of pressure.*
- ☒ *through its centre of gravity.*
- ☐ *through a point defined as the datum point.*

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15

(Please view annex 031-038 rev. 05.11.2004)
For the following boom to be in balance:

See Attachments:

1. 031-038.gif

Annex

(Attachment
No.1)

☒ $A = B \times Fb / Fa$

☐ $A = B \times Fa / Fb$

☒ $A = B (Fa + Fb)$

☐ $A = B + Fb / Fa$

16

The best lift/drag ratio of a glider occurs when parasite drag is

- ☒ equal to total drag.
- ☐ less than induced drag.
- ☐ equal to induced drag.

17

During recoveries from unusual attitudes, level flight is attained the moment

- ☐ a zero rate of climb is indicated on the VSI.
- ☐ the horizon bar on the attitude indicator is exactly overlapped with the miniature airplane.
- ☒ the altimeter and airspeed needles stop prior to reversing their direction of movement.

18

Which statement is correct for a descent without engine thrust at maximum lift to drag ratio speed?

- ☐ The higher the gross mass the lower is the speed for descent.
- ☐ The mass of an aeroplane does not have any effect on the speed for descent.
- ☒ The higher the gross mass the greater is the speed for descent.
- ☐ The higher the average temperature (OAT) the lower is the speed for descent.

19

How does the thrust of fixed propeller vary during take- off run ? The thrust

- ☐ decreases slightly while the aeroplane speed builds up.
- ☒ increases slightly while the aeroplane speed builds up.
- ☐ has no change during take-off and climb.
- ☐ varies with mass changes only.

20

What effect has a downhill slope on the take-off speeds?The slope

- ☐ decreases the take-off speed V_1 .
- ☒ increases the IAS for take-off.
- ☐ decreases the TAS for take-off.
- ☐ has no effect on the take-off speed V_1 .

21

The coefficient of lift can be increased either by flap extension or by

- ☐ decreasing the 'nose-up' elevator trim setting.
- ☐ increasing the CAS.
- ☐ increasing the TAS.
- ☒ increasing the angle of attack.

22

For a turboprop powered aeroplane, a 2200 m long runway at the destination aerodrome is expected to be "wet". The "dry runway" landing distance, should not exceed:

☐ 1540 m.

☒ 1339 m.

☐ 1771 m.

☐ 1147 m.

23

How is V2 affected if take off flaps 20° is chosen instead of take off flaps 10°?

- ☐ V2 has no connection with T/O flap setting, as it is a function of runway length only.
- ☐ V2 has the same value in both cases.
- ☐ V2 increases in proportion to the angle at which the flaps are set.
- ☒ V2 decreases if not restricted by VMCA.

24

The best rate of climb at a constant gross mass

- ☐ decreases with increasing altitude since the thrust available decreases due to the lower air density.
- ☒ increases with increasing altitude due to the higher true airspeed.
- ☐ is independent of altitude.
- ☐ increases with increasing altitude since the drag decreases due to the lower air density.

25

Which of the following statements is correct?

- ☐ Induced drag decreases with increasing angle of attack.
- ☒ Induced drag decreases with increasing speed.
- ☐ Induced drag is independant of the speed.
- ☐ Induced drag increases with increasing speed.

26

For a piston engined aeroplane, the speed for maximum range is :

- ☒ that which gives the maximum value of lift
- ☐ that which gives the minimum value of drag.
- ☐ that which gives the maximum lift to drag ratio.
- ☐ 1.4 times the stall speed in clean configuration.

27

A constant headwind component

- ☐ increases the maximum endurance.
- ☐ increases the angle of flight path during climb.
- ☐ decreases the angle of climb.
- ☒ increases the best rate of climb.

28

The optimum cruise altitude increases

- ☒ if the aeroplane mass is increased.
- ☐ if the aeroplane mass is decreased.
- ☐ if the tailwind component is decreased.
- ☐ if the temperature (OAT) is increased.

29

The induced drag of an aeroplane

- ☐ increases with increasing airspeed.
- ☐ decreases with increasing gross weight.
- ☒ decreases with increasing airspeed.
- ☐ is independent of the airspeed.

30

Which statement is correct for a descent without engine thrust at maximum lift to drag ratio speed?

- ☐ A tailwind component decreases the ground distance.
- ☐ A headwind component increases the ground distance.
- ☐ A tailwind component increases fuel and time to descent.
- ☒ A tailwind component increases the ground distance.

31

If an aircraft has a transponder, encoding altimeter, and DME, the proper equipment suffix to be entered on a flight plan is

☒ R.

☐ U.

☐ A.

32

GIVEN:

Usable fuel at takeoff 40 gal, Fuel consumption rate 12.2 gal/hr Constant
groundspeed 120 kts
Flight time since takeoff 1 hr 30 min

☐ 156 NM.

☐ 121 NM.

☒ 216 NM.

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33

GIVEN:

Usable fuel at takeoff 40 gal
Fuel consumption rate 12.2 gal/hr
Constant groundspeed 120 kts
Flight time since takeoff 1 hr 30 min

☐ 215 NM.

☐ 121 NM.

☒ 176 NM.

34

How is your flight plan closed when your destination airport has IFR conditions and there is no control tower or Flight control Center(FCC) on the field?

- ☐ The ARTCC controller will close your flight plan when you report the runway in sight.
- ☒ Upon landing, you must close your flight plan by radio or by telephone to any FCC or ATC facility.
- ☐ You may close your flight plan any time after starting the approach by contacting any FCC or ATC facility.

35

How long will a Flight Service Station hold a VFR flight plan past the proposed departure time?

☐ 1 hour.

☒ 30 minutes.

☐ 2 hours.

36

All fuel tanks should be fired during preflight to determine

- ☐ if there are any leaks in the tanks.
- ☒ if the pilot light functions properly on each tank.
- ☐ burner pressure and condition of the valves.

37

From which of the following would you expect to find information regarding known short unserviceability of VOR, TACAN, and NDB ?

☐ D ATCC broadcasts

☒ A NOTAM

☐ C SIGMET

☒ B AIP (Aeronautical Information Publication)

38

Given: true course (TC) 017; W/V 340/30; TAS 420 kt
Find: wind correction angle (WCA) and ground speed (GS)

☒ WCA +2° ; GS 396 kt

☐ WCA +2° ; GS 416 kt

☐ WCA -2° ; GS 426 kt

☐ WCA -2° ; GS 396 kt

39

In an ATS flight plan, Item 15 (route), a cruising pressure altitude of 32000 feet would be entered as :

☒ A FL320

☐ C 32000

☐ D F320

☐ B S3200

40

A "current flight plan" is a:

- ☐ *filed flight plan.*
- ☐ *flight plan in the course of which radio communication should be practised between aeroplane and ATC.*
- ☐ *filed flight plan with amendments and clearance included.*
- ☒ *flight plan with the correct time of departure.*

Attachment No. 1

Questions 15

Annex

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